

Southeast University Examination Paper (MidTerm Self-test)

Course Name Principles of Compiling Examination Term 15-16-2 Score _____Related Major _____ Examination Form Close test Test Duration 120 Mins

There are 5 problems in this paper. You can write the answers in English or Chinese.

1. Please construct **context-free grammars without ϵ -productions** for the following language.

$L = \{ \omega \mid \omega \in (a,b,c,d)^* \text{ and the numbers of } a\text{'s and } b\text{'s and } c\text{'s occurred in } \omega \text{ are } \mathbf{even}, \text{ and } \omega \mathbf{starts with a or b} \}$

2. Please construct a **DFA with minimum states** for the following regular expression. (15%)

$a(a|b)^*(a|b)^*(a|b)b$

3. Please **eliminate the left recursions (if there are)** and **extract maximum common left factors (if there are)** from the following context free grammar, and then decide **the resulted grammar** is whether a LL(1) grammar by **constructing the related LL(1) parsing table.**(15%)

$S \rightarrow \text{begin } L \text{ end} \mid \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid \text{while } E \text{ do } S \mid a$

$L \rightarrow L; S \mid S$

$E \rightarrow E \text{ or } F|F$

$F \rightarrow F \text{ and } G|G$

$G \rightarrow (E)|b$

4. Please **construct a LR(1) parsing table for the following ambiguous grammar with your own defined additional conditions (You determine the required additional conditions by yourself).**(15%)

$S \rightarrow \text{if } E \text{ then } S | \text{if } E \text{ then } S \text{ else } S|a$

$E \rightarrow E \text{ and } E | E \text{ or } E|(E)|b$

5. Please construct **an annotated parse tree** for the input string $4*5+6$ where the syntax-directed definition is as following (10%):

Productions	Semantic Rules
$E \rightarrow E_1 * T$	$E.val = E_1.val * T.val$
$E \rightarrow T$	$E.val = T.val$
$T \rightarrow T_1 + F$	$T.val = T_1.val + F.val$
$T \rightarrow F$	$T.val = F.val$
$F \rightarrow i$	$F.val = i.\text{lexval}$

